MA1521 L1 Week 08 Wed 20261007

Evaluate the double integral

$$\iint_R \frac{dA}{r}; \qquad R \text{ is the region inside the first-quadrant loop of } r = \sin 2\theta.$$

Evaluate the double integral

$$\iint_R \tan^2\theta \, dA; \qquad R \text{ is the triangle with vertices at } (0,0),\ (1,0),\ (1,1).$$

Evaluate the double integral

$$\iint_R \frac{dx\,dy}{1+x^2+y^2}; \qquad R \text{ is the first-quadrant portion of the interior of } x^2+y^2=a^2$$

Evaluate the double integral

$$\iint_R \frac{dx\,dy}{\sqrt{1-x^2-y^2}};$$

R is the right half-disk of radius $\frac{1}{2}$ centered at $(0, \frac{1}{2})$.

Evaluate $\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \ln\left(x^2 + y^2 + 1\right) dx\, dy$